

10.2 Outliers

Lesson Introduction

 Benchmarks	MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation. MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.		 Learning Targets	- I can determine the implications an outlier has in a given data set. - I can relate the impact of an outlier to which measures of center and variation best describe a distribution.
 Benchmark Coherence	Building On MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		Working Toward MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.	
 Essential Questions	- What is an outlier? - How does an outlier affect measures of center and variation? - Are there times when an outlier does not affect the measures? - What measures of center and variation can be used to describe sets of numerical data? - How does removing an outlier from a data set affect the measures of center and variation?			
 Lesson Narrative	This lesson begins with students exploring data presented in a line plot. Students analyze the information presented and apply what they know about measures of center and variation to describe and summarize the data, considering the impact an outlier has on the data set and what happens once it is removed. Students summarize the impact of outliers on measures of center and variation, applying this understanding to synthesize data presented in a histogram and other data displays.			
 Component Summary	10.2.1 Warm-Up (~5 min.)	Watch a video and reflect on the impact of errors (<i>SE p. 109</i>)	10.2.4 Guided Practice (10 min.)	Calculate impact of outlier on data set (<i>SE p. 112</i>)
	10.2.2 Exploration (10 min.)	Work with a partner to find the five-number summary and IQR (<i>SE p. 110</i>)	10.2.5 Exploration (10-15 min.)	Discover how to find best fit measurement of variation and center (<i>SE p. 113</i>)
	10.2.3 Bring It Together (5 min.)	Synthesize how outliers affect data (<i>SE p. 111</i>)	10.2.6 Your Turn! (5 min.)	Analyze how an outlier impacts a data set (<i>SE p. 115</i>)
	10.2.7 Wrap-Up (~5 min.)	Reflect on understanding of outlier impact on data (<i>SE p. 115</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Outlier - Interquartile range		(Optional) Paper (Optional) Tape and labels for dot plot (Optional) Computer	
	Additional Preparation 10.2.1 Warm-Up contains a video to accompany the question prompts.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly determine an outlier can only be higher or only be lower than the rest of the data. - Students may think that an outlier will always significantly change the mean. - Students may struggle to draw conclusions about the effect of outliers on measures of center and variation. - Students may think that removing outliers will always have the same impact on the data set, regardless of whether the outlier falls above or below the norm in the set.	- Consider providing examples and nonexamples of data sets containing outliers to students to decipher the differences. - Allowing students to use calculators during the lesson will help students focus on the effect of outliers on measures of center and variation. Consider have no-calculator questions where students turn their calculator over. - If time is an issue, then consider assigning 10.2.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 10.2.4 Guided Practice provides an opportunity for students to engage in a Think-Pair-Share. This routine encourages students to synthesize information as they first consider the question and then discuss with a partner to build onto or revise their thinking. Through this routine, students have the chance to deepen understanding of the impact of outliers.	MA.K12.MTR.7.1: Real-World Contexts Students will be making connections between data displayed on graphs and histograms and real-world contexts throughout this lesson. Students will use the information presented to determine measures of center and variation and connect this back to the context of the problems to reason about the impact of outliers when summarizing data distributions.
 Differentiate Instruction	Utilizing various display methods to show the data provides a variety of visual entry points for learners. Students can engage with the content through histograms, dot plots, and tables. Encourage students to create their own data displays to help answer questions about the data if you notice them struggling to conceptualize the data presented without a visual display. Students work collaboratively throughout the exploration and guided practice, providing opportunity for mathematical discourse as well as auditory entry points.	

Lesson Notes:

10.2.1 Warm-Up: Precision Is Key

Component Overview: Play the Career Profile video on the quality control engineer for the entire class. After students watch the video, prompt them to answer the two questions independently. Consider stopping the video at minute 2:22 to allow students enough time to reflect on and complete the questions.



10.2.1 Warm-Up: Precision Is Key

Becky Miller is a quality control engineer with GE Aviation. Like many other careers, manufacturing engineering has many moving parts that make up an entire process. Becky says that manufacturing is problem-solving.

1. List at least two steps that you take when faced with a difficult task.

Answers will vary. Understand the problem. Make a plan. Keep trying.

Becky mentions that an error “about the size of a human hair” can make a difference in manufacturing jet engines.

2. What is a time that you have made a small error that caused a big impact in your next steps?

Answers will vary. I used the wrong inverse operation when solving an equation. I woke up late for school.



This Warm-Up positions students to consider the importance of precision when working with numbers in mathematics. To connect to the rest of the lesson, ask questions like:

- *Would rounding numbers to the wrong decimal place ever impact the correctness of your answer?*
- *When thinking about answers would you ever say that 141.3 phone calls were made?*
- *When speaking of money is it accurate to write \$5.6?*

10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Component Overview: Students will work with a partner to calculate the five-number summary and IQR of a data set. Students then remove the outliers from the set and recalculate to find the effects an outlier has on the values used to summarize the distribution. Encourage students to make connections between measures that change and do not change when the outlier is removed.

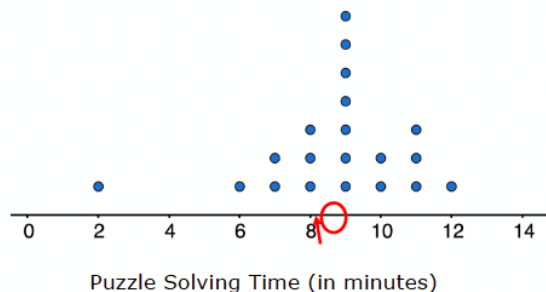


10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically

Work with your partner to complete the following.

- Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a dot plot displaying the data.

2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12



- Describe the distribution of data.
Answers will vary. The data is symmetrical if not for the outlier with the value of 2. The outlier causes the data to skew to the left.
- Determine the mean of the data set. Then, draw an arrow (↑) below the number line that points to this value.
8.7
- Determine the median of the data set. Then, circle the value on the number line that represents this value.
9
- Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.
Answers will vary. The mean and median are very close to each other. The mean is slightly less than the median.



While students are exploring in small groups or pairs, monitor for the use of both informal and formal language in their descriptions. Use their responses as probing questions for 10.2.3 Bring It Together for meaningful synthesis.

For example, you may notice Student A describing the distribution as “normal-looking” or using similar informal language, while Student B in another group may describe it as symmetrical.

As you take note of student responses here, see possible question stems to include in 10.2.3 Bring It Together.



Have students make a connection between the outlier and the median and mean.

10.2.2 Exploration: Determining, Describing, and Comparing Numerical Data Represented Graphically (continued)

e. What is the range of the data?

10

f. Find the interquartile range (IQR).

Q1	Q3	IQR
8	10	2

2. Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.

Answers will vary. Removing the fastest time will make the graph a more accurate description of the time it takes to complete the puzzle.

3. How does the shape of the graph change after 2 is removed?

Answers will vary. The graph would be more symmetrical. The graph will be more representative of the time it takes to complete the puzzle.

4. Compare the new values with the original values.

	Original Value	>, <, =	New Value after 2 is Removed
Mean	8.7	<	9.1
Median	9	=	9
Range	10	>	6
IQR	2	=	2

5. Which measure(s) of center will be affected if the outlier is removed?

- ☐ mean
- ☐ median
- ☐ both the mean and median

6. Which measure(s) of variation will be affected if the outlier is removed?

- ☐ range
- ☐ interquartile range
- ☐ both the range and the interquartile range



Notice and Wonder

Implement a Notice and Wonder routine to encourage students to notice what happens to each measure of center and variation after the outliers have been removed.

- What do you notice about the original values compared to the new value of each category after 2 is removed?
- What do you notice about the two categories that stayed the same?
- What do you wonder? Do you think this happens when outliers are removed from other sets of data?

10.2.3 Bring It Together: Outlier Effects

Component Overview: Students complete statements to describe how the presence of outliers impact the measures of center and variation.



10.2.3 Bring It Together: Outlier Effects

1. Complete each statement to describe the effect the outlier, 2, has on each measure of center or variation.

- The presence of an outlier does not have a significant effect on the ☐ mean.
☒ median.
- The presence of an outlier will change, often significantly, the value of the ☒ mean.
☐ median.
- The presence of an outlier does not have a significant effect on the ☒ IQR.
☐ range.
- The presence of an outlier will change, often significantly, the value of the ☐ IQR.
☒ range.



When outliers are present or a distribution is skewed, the **median** is the preferred measure of center and the **IQR** is the preferred measure of variation to describe the distribution.



Use inquiry and talk moves to have students make sense of their learning with the sentence frames and the blue box, using questions to have them formalize their descriptions from the Exploration.

- Student A, you described the data distribution as 'normal-looking' – can you say more about what you mean by that?
- Student B, how did you describe the distribution?
- Student A, based on Student B's response, would you change your answer to the sentence frame? Why or why not?
- Student B, what impact does the outlier have on the distribution?
- Student A, based on that response, should we use median or IQR in the Exploration? Why?

10.2.4 Guided Practice: Data Outliers

Component Overview: In this component, students work with a partner to apply their knowledge of calculating the five-number summary and IQR to determine how an outlier can impact the data. Students remove an outlier to determine how it impacts a data set and the measures of center and variation.



10.2.4 Guided Practice: Data Outliers

1. Zoe needed exact change to purchase a snack and a drink from the vending machine. She asked her friends how much change, in dollars, they had and listed the results.

0.15, 0.15, 0.20, 0.25, 0.25, 0.30, 0.30, 0.30, 0.35, 0.75, 0.85

- a. What do you notice about the data?
Answers will vary. All numbers are less than \$1.
- b. Will the mean be higher, lower, or approximately the same as the median? Explain your reasoning or justify your answer by showing your work.
Answers may vary. Slightly higher. Mean is 0.35 and the median is 0.30.
- c. Describe the impact of removing 0.85 from the data set.
Answers may vary. The data becomes more symmetrical. Now 0.75 is the only outlier. The mean will decrease.
- d. The values 0.85 and 0.75 are outliers in this data set. What happens to the range if 0.85 and 0.75 were both removed from the data set?
The range becomes smaller and the spread of the data is smaller
- e. The IQR of the full data set is 0.15.

The IQR with 0.85 and 0.75 removed is 0.125.
- f. Is there a big difference between the IQR of the original data set and the modified data set?
No, the IQR is not significantly affected by the removal of outliers



Consider providing a graphic organizer or table for students to use when finding the mean, median, and interquartile range.



To incorporate digital learning, allow students to type the numbers into a column on an Excel spreadsheet to manipulate the data and view how changing the different results changes the data.



Think-Pair-Share
Engage students in a Think-Pair-Share routine to complete this problem. This will provide students with the opportunity to formalize their thinking and synthesize the impact of outliers in a data set and on measures of center and variation.

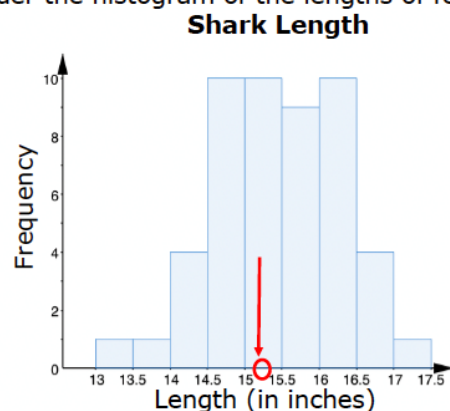
10.2.5 Exploration: Measures of Center

Component Overview: In this exploration, students work with a partner to determine the measures of center and spread that best describe the data presented in two different histograms. The first histogram is relatively symmetrical, while the second is skewed to the right. Students analyze the shape of the histograms to draw conclusions about how outliers impact the data sets. Students will also evaluate the best measures of center and spread to use when describing symmetrical versus skewed data sets.



10.2.5 Exploration: Measures of Center

1. Consider the histogram of the lengths of female sharks.



- a. Describe the shape of the histogram.
The graph is symmetrical, with most sharks having a length between 14.5 and 16.5 inches.
- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of the data.
Shown on graph.
- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.
Shown on graph.
- d. Compare your answers to B and C with your partner's answers. Discuss your reasoning, and if necessary, make changes.



The histogram provides a visual entry point. Consider having students further distinguish between median and mean by using different colored pencils to label each on the graph.



In a symmetrical graph, the mean and the median are located in the middle of the graph, approximately in the same location.

10.2.5 Exploration: Measures of Center (continued)

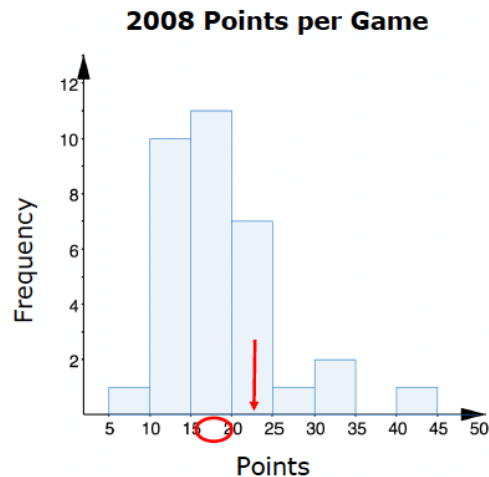
- e. The mean and the median of this histogram are located between 15 – 15.5. Discuss with your partner which measure most accurately describes the center of the histogram. Summarize your discussion.

Answers will vary. Both measures have about the same value, but the mean best describes the data because the average represents a typical length.



In a symmetrical graph with no outliers, the **mean** is the preferred measure of center and the **range** is the preferred measure of variation to describe the distribution.

2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.



- a. Describe the shape of the histogram.
Skewed to the right.



To solidify conceptual understanding, read all three blue boxes in 10.2.5 as a class and ask the following questions:



- Are the values for mean and median usually close together?
- What could cause them to be farther apart?
- Why is the mean the preferred measure of center when a data set has no outliers?
- Why is the range the preferred measure of variation to use with mean?



Consider allowing students to draw pictures to match the words in the second blue box for those who would benefit from a visual of the statement.

10.2.5 Exploration: Measures of Center (continued)

- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of this data.

Shown on graph.

- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.

Shown on graph.

- d. In a symmetrical graph, the mean is the preferred measure of center to use. The histogram for points scored is skewed. Discuss with your partner why the mean would not be a good representation of the center for points scored. Summarize your discussion.

Answers may vary. The value of the mean is affected by the higher values that are causing the shape to be skewed. Using the mean may falsely represent the center as higher than it actually is.

- e. In this graph, the median is located between 15 – 20 and the mean is in the 20 – 25 interval. Which measure most accurately describes the center of the histogram?

The median because the peak and center of the data is closer to 15-20.

- f. Which measure of variation should be used to describe the distribution?

Answers will vary. The range is affected by the few large values that are causing the distribution to be skewed to the right, but they do not affect the IQR. The IQR should be used to describe the variation.



In a skewed graph or a graph with outliers, the **median** is the preferred measure of center and the **IQR** is the preferred measure of variation to describe the distribution.



For a digital learning extension, consider allowing students to enter each number into an Excel spreadsheet and create a chart with the data. Students can then find the mean and median of the given data using the equation functions in Excel.



Students may not understand why the mean would not be a good representation of the center in a skewed graph. To further scaffold this section, ask guiding questions if needed:

- What do you know about outliers? How do they impact data distributions?
- Is the mean impacted by high/low outliers? How?
- Is the median impacted by outliers? Why or why not?



Note that the values of the IQR and range cannot be determined since only the histogram is given. A box plot or data set would need to be included to determine those exact values.

10.2.6 Your Turn!

Component Overview: Students explain which value in a data set they think could be an outlier before describing the effect of removing it from the data set.



10.2.6 Your Turn!

The following data set shows the number of tickets a movie theater sold in a day for each movie.

162	623	76	277	181	64	105
265	82	263	124	167	150	245

- One of the values in the data set is an outlier. Explain which value you think is the outlier.

Answers will vary. 623

This value is much larger than the rest of the data set.

- Complete the statement.

Removing the outlier, 623, causes the mean

to
☐ increase
☒ decrease
☐ remain the same
 and the range to
☐ increase
☒ decrease
☐ remain the same

 because it is much
☒ larger
☐ smaller
 than the other values in the data set.



To challenge students and further build conceptual understanding, ask questions like:

- What are some reasons statisticians may remove an outlier?*
- What are some possible reasons outliers could have occurred in this context?*

10.2.7 Wrap-Up: Cool Down

Component Overview: This Wrap-Up assesses students' abilities to determine if and how an outlier affects different measures of center and variation for a set of data.



10.2.7 Wrap-Up: Cool Down

1. The presence of an outlier has a significant effect on which measures of center and spread? Select all that apply.

- ☒ Mean
- ☐ Median
- ☐ Interquartile Range
- ☒ Range

2. A data set has an outlier that is significantly less than most of the other values.

- a. Explain how this impacts the measures selected in number 1.

A lower outlier will cause the mean to be smaller than the median and will increase the range of the data.

- b. Which measures of center and variation would you use to best describe the data?

Answers may vary. Median and IQR (interquartile range).



Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson and inform groupings for remediation or partner work in the next lesson.